

Project Design Phase-II
Solution Requirements (Functional & Non-functional)

Date	18 June 2026
Team ID	
Project Name	ShopEZ
Maximum Marks	4 Marks

Functional Requirements:

Following are the functional requirements of the proposed solution.

FR No.	Functional Requirement (Epic)	Sub Requirement (Story / Sub-Task)
FR-1	Authentication & Security	Registration via email/password Login via JWT validation Role-based route protection.
FR-2	Adaptive Product Catalog	Dynamic UI sizing based on category Similar product recommendations.
FR-3	Cart & Ordering	Local storage cart integration Order creation with shipping details
FR-4	Admin Cockpit	Metrics aggregator for dashboard Product metadata CRUD control
FR-5	Reporting & Invoicing	Export orders to CSV Print-specific invoice styling

Non-functional Requirements:

Following are the non-functional requirements of the proposed solution.

FR No.	Non-Functional Requirement	Description
NFR-1	Usability	Storefront must be 100% fluid across screen widths using Tailwind responsive utilities and adapt layouts to reduce cognitive load.
NFR-2	Security	Passwords must be hashed using bcrypt, and admin API endpoints must validate JWT roles, returning 403 errors on violations.
NFR-3	Reliability	Database inputs require Mongoose validations, and the API must return structured JSON 500 errors instead of crashing during failures.
NFR-4	Performance	Backend API response times must stay under 500ms, and frontend static asset bundles must be optimized via tree-shaking to remain under 300KB.

NFR-5	Availability	The system must ensure 99.9% uptime (high availability) by utilizing managed cloud infrastructure (e.g., Vercel for the React frontend, MongoDB Atlas for the database) that supports automated failovers and zero-downtime deployments.
NFR-6	Scalability	Architecture must remain decoupled, and stateless API execution must be supported to allow server containers behind a proxy load balancer.